container, successfully detecting an open port 80 on Metasploitable, which highlights the ability to perform realistic penetration testing scenarios entirely within the testbed. In addition to the detection of open port in the mentioned container, this scanner finds the open ports in the 4 IPs as mentioned by the CIDR notation 4. Of the 4 IPs available in the subnet, 2 IPs have TCP open ports, which are a perfect entry point for security researchers to enter the system for further security experiments.

Running a security testbed using Docker containers on Windows with WSL 2 offers significant advantages over traditional virtual machine environments. This setup is highly efficient, allowing multiple machines to run smoothly with a much smaller memory footprint and faster startup times, thanks to containers sharing the host OS kernel rather than each requiring a full operating system. As a result, you can deploy and manage many interconnected security lab environments simultaneously without the heavy resource consumption and sluggish performance often associated with virtual machines. This makes Docker-based testbeds not only more scalable and cost-effective, but also ideal for dynamic cybersecurity experimentation and training.

You can run multiple security-focused machines in this setup—like Kali, DVWA, Metasploitable, and Snort—as interconnected Docker containers, all within an isolated virtual network. One standout advantage is that while these containers can freely communicate with each other for realistic attack and defence scenarios, they are not accessible from the Windows host machine. This dramatically reduces the risk to your primary operating system, ensuring that any vulnerabilities or exploits tested within the containers remain safely contained. Additional benefits include easy setup and management, reproducibility of your lab environment, and the performance enhancements provided by WSL 2's native Linux kernel support. Overall, this approach offers a safe, efficient, and highly customisable platform for hands-on security experimentation without compromising your host system. END

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